Indian Institute of Technology Delhi

Summary Undergraduate Research Award (SURA)

Analysing Multiphase Flows through Catalytic Converters using Graph Neural Networks

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1 Introduction

Catalytic converters are integral components of modern vehicles, serving as critical emissions control devices. These ingenious devices play a pivotal role in reducing harmful pollutants emitted from internal combustion engines into the atmosphere, thereby mitigating environmental pollution and safeguarding public health. Understanding their chemical, mechanical, and flow properties is essential to grasp their significance in environmental protection.
A multiphase reactor is a type of chemical reactor in which more than one phase (solid, liquid, or gas) is present during the reaction process. These reactors are used in various industries for e.g:- energy for different purposes due to their ability to handle complex reactions involving multiple phases simultaneously.
Machine Learning is a valuable toolkit that complements incomplete domain-specific knowledge in conventional experimental and computational methods. Machine Learning can provide flexible techniques to facilitate the conceptual development of new robust predictive models like:-GNNs for multiphase flows and reactors by finding hidden pattern/information/mechanism in a data set. We thereby comprehensively analyze multiphase flow systems using such predictive models.

2 Objectives

The Primary objective of the research is the analysis of small and large-scale simulated flow behavior in catalytic converters and industrial multiphase reactors by training a Graph Neural Network Model on small scale multiphase flows through catalytic converters and using the trained model to find patterns/information in large scale multiphase flows.

3 Concepts

We briefly describe the main components of multiphase flows through catalytic converters.

1. **Catalytic Converters**
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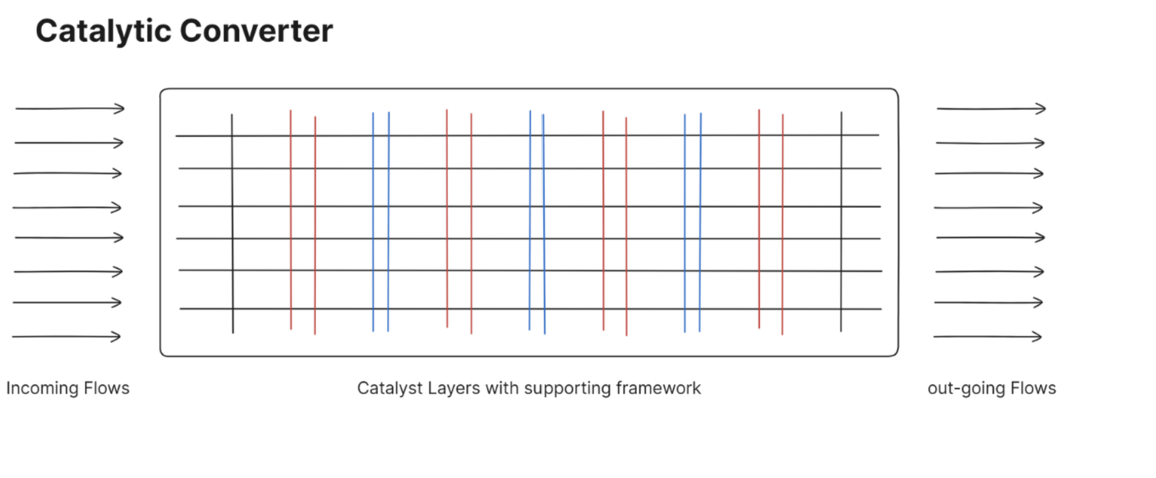


Figure 1: Flows through Catalytic Converters

2. **Chemical Processes**
   At the heart of catalytic converters lie intricate chemical reactions facilitated by catalysts, typically composed of precious metals like platinum, palladium, and rhodium. These metals serve as active sites where pollutants undergo transformation. When exhaust gases containing unburned hydrocarbons (HC), carbon monoxide (CO), and nitrogen oxides (NOx) pass over the catalyst surface, a series of chemical reactions occur.

3. **Mechanical Processes**
   Beyond the chemical reactions, catalytic converters exhibit crucial mechanical properties to ensure optimal performance. The physical structure of the converter is designed to maximize the contact between exhaust gases and the catalyst surface. Honeycomb-like structures made of ceramic or metallic substrates provide a large surface area for catalytic reactions to take place.

4. **Graph Neural Networks**
   Graph Neural Networks (GNNs) have gained significant importance in various fields of science and engineering, including multiphase flow analysis. Multiphase flow refers to the movement of multiple phases of matter, such as gas, liquid, and solid, within a system.Here's why Graph Neural Networks are particularly important for multiphase flow analysis:

   (a) **Complexity of Multiphase Flows**
   Multiphase flow systems are inherently complex, with interactions occurring between different phases and various spatial and temporal scales. Traditional computational fluid dynamics (CFD) methods often struggle to capture these complexities accurately due to their reliance on structured grids or meshes. GNNs offer a promising alternative by allowing for the representation of complex multiphase flow systems as graphs, where nodes represent entities such as particles or cells, and edges represent interactions between them.

   (b) **Learning Complex Dynamics**
   GNNs are capable of learning complex dynamics and patterns directly from data. In multiphase flow analysis, this means that GNNs can learn the behavior of different phases, interfaces, and interactions without relying heavily on handcrafted features or domain-specific knowledge. This data-driven approach allows for more accurate modeling and prediction of multiphase flow phenomena.

   (c) **Scalability and Efficiency**
   Traditional CFD methods often face scalability issues when dealing with large-scale multiphase flow systems or complex geometries. GNNs offer scalability and efficiency advantages, as they can process graph-structured data in a parallel and distributed manner, making them suitable for handling large-scale multiphase flow simulations.

   (d) **Integration with Simulated and Experiment data**
   GNNs can be integrated seamlessly with simulation data generated from CFD simulations or experimental data obtained from laboratory measurements. By combining GNN-based approaches with traditional simulation techniques, we can leverage the strengths of both methods to enhance the accuracy and efficiency of multiphase flow analysis.

These points also serve as the **MOTIVATION** behind this project.

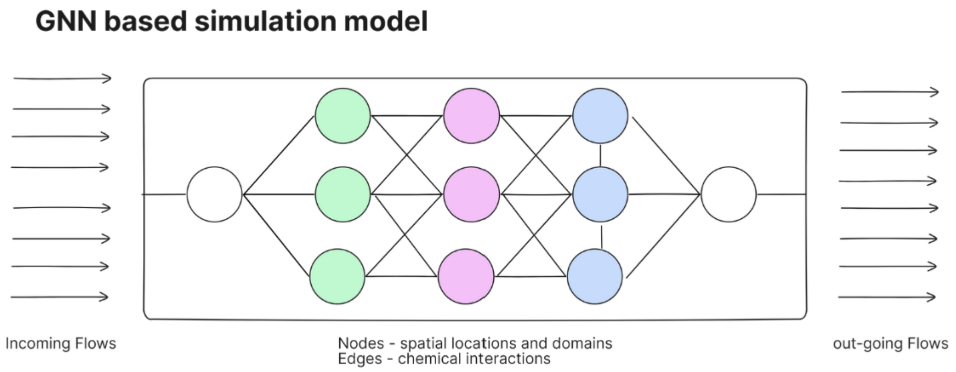


Figure 2: GNN based Simulation Model

4 Approach

1. **Simulation of Multiphase Flows through Catalytic Converters**
   The first phase of study will gravitate towards simulating the multiphase flows through Catalytic converters that we wish to study. These converters rely on intricate flows of different phases through densely packed catalyst particles to achieve optimal conversion rates.

2. **Representation of the Simulated Data as Graphs for training a Graph Neural Network Model**
   Multiphase flow domains can be represented as graphs, where nodes correspond to discrete spatial locations or computational cells, and edges denote interactions between neighboring nodes.GNN architectures are designed to learn and infer the behavior from the graph-structured representation of the simulated multiphase flow data. Through iterative training, GNNs adaptively learn the underlying relationships and dependencies within the flow system.The performance of GNN-based models is assessed through validation procedures, including comparison against experimental data, traditional computational methods, or analytical solutions.

3. **Optimization of the GNN Model**
   The third phase will be focused on the study of different optimizations techniques to improve both the precision and the accuracy of the GNN model which will be trained on our simulated data.This model will also serve as the basis for construction and analysis of different architectures of catalytic converters to improve their operational efficiency.

4. **Extrapolation to Large Scale Flows in Industrial Reactors**
   Later we would like to extend the study to multiphase flows on large scale reactors. It's speculative that extrapolation of the small scale simulated flow data would lead to compromise in accuracy and precision or spike in computational power requirement making it impractical for design and study purpose, therefore the third phase of this project will be about counterbalancing the loss of accuracy and computational intensity of this GNN model for achieving results for large scale multiphase flows through industrial reactors.

5 Research Methodology

- Test and validate the trained Graph Neural Network on larger domains. (GNNs are known to predict on scaled-up domains).
- Use appropriate time scales to train GNN using simulation data:

$$\tau_1 = \frac{L}{V}\frac{1}{Re \cdot Ca} \quad \text{Inertial time scale} \tag{1}$$

$$\tau_2 = \frac{L \cdot Ca}{V} \quad \text{Capillary time scale} \tag{2}$$

- Demonstrate if the GNN can predict long-term temporal evolution of the two-phase interface.
- Use an analogous system of force chain prediction to help and understand for ourselves about GCNs.
- Build our own GNN models that produce satisfactory results on force chain predictions.

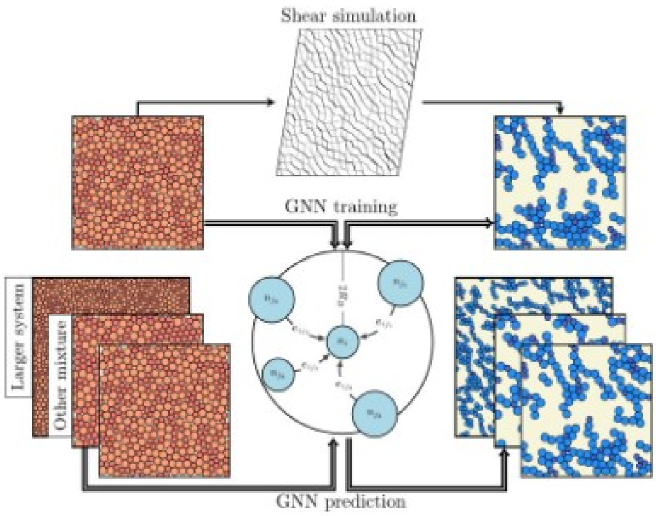
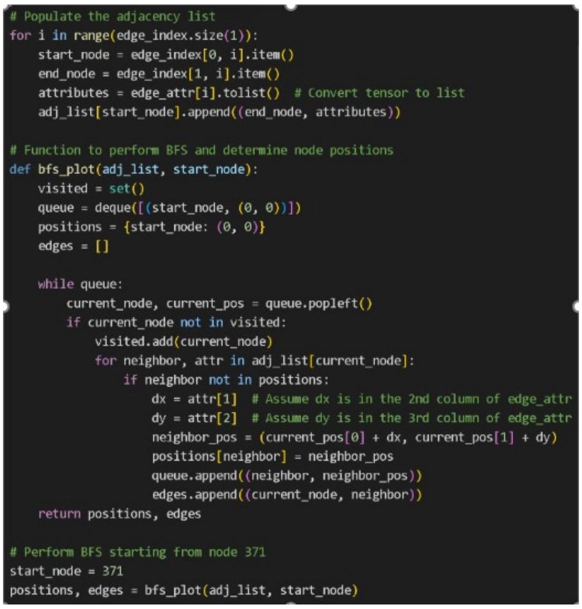


Figure 3: Robust prediction of force chains in jammed solids using graph neural networks

- Instead of simulating large-scale multiphase flows, which take months and significant computational costs, we will simulate multiphase flows through small-scale packed beds.
- The robustness and scalability of the Graph Neural Network model will allow us to extend this model trained on small-scale flows to larger-scale multiphase flows through multiphase reactors.
- Instead of training our Graph Neural Network at stages of stress and non-stress in the multiphase flow particles, we will train the model at different temporal stages of the multiphase flow through overlapping sub-regions.

6 Visualizing our data

- We have used BFS algorithm for visualizing the arrangement of nodes in the graph. The size of the nodes depends on the node features and the color depends on whether they are part of force chains or not.



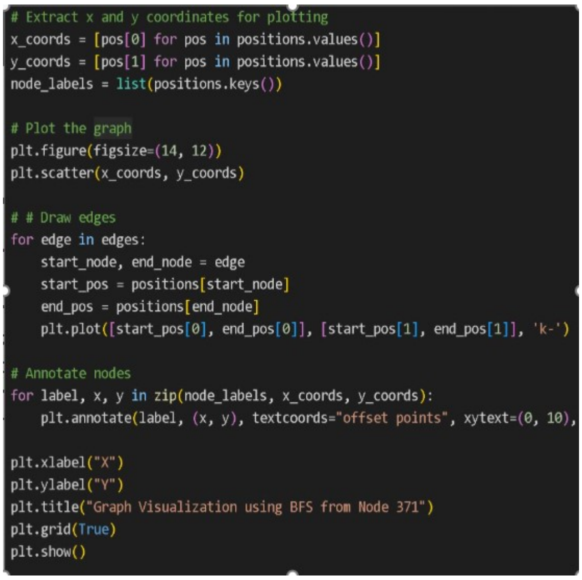
```python
# Populate the adjacency list
for i in range(edge_index.size(1)):
    start_node = edge_index[0, i].item()
    end_node = edge_index[1, i].item()
    attributes = edge_attr[i].tolist()  # Convert tensor to list
    adj_list[start_node].append((end_node, attributes))

# Function to perform BFS and determine node positions
def bfs_plot(adj_list, start_node):
    visited = set()
    queue = deque([(start_node, (0, 0))])
    positions = {start_node: (0, 0)}
    edges = []

    while queue:
        current_node, current_pos = queue.popleft()
        if current_node not in visited:
            visited.add(current_node)
            for neighbor, attr in adj_list[current_node]:
                if neighbor not in positions:
                    dx = attr[1]  # Assume dx is in the 2nd column of edge_attr
                    dy = attr[2]  # Assume dy is in the 3rd column of edge_attr
                    neighbor_pos = (current_pos[0] + dx, current_pos[1] + dy)
                    positions[neighbor] = neighbor_pos
                    queue.append((neighbor, neighbor_pos))
                    edges.append((current_node, neighbor))
    return positions, edges

# Perform BFS starting from node 371
start_node = 371
positions, edges = bfs_plot(adj_list, start_node)
```

(a) Code Snippet 1



```python
# Extract x and y coordinates for plotting
x_coords = [pos[0] for pos in positions.values()]
y_coords = [pos[1] for pos in positions.values()]
node_labels = list(positions.keys())

# Plot the graph
plt.figure(figsize=(14, 12))
plt.scatter(x_coords, y_coords)

# # Draw edges
for edge in edges:
    start_node, end_node = edge
    start_pos = positions[start_node]
    end_pos = positions[end_node]
    plt.plot([start_pos[0], end_pos[0]], [start_pos[1], end_pos[1]], 'k-')

# Annotate nodes
for label, x, y in zip(node_labels, x_coords, y_coords):
    plt.annotate(label, (x, y), textcoords="offset points", xytext=(0, 10),

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Graph Visualization using BFS from Node 371")
plt.grid(True)
plt.show()
```

(b) Code Snippet 2

Figure 4: Code Snippets for data visualization using BFS

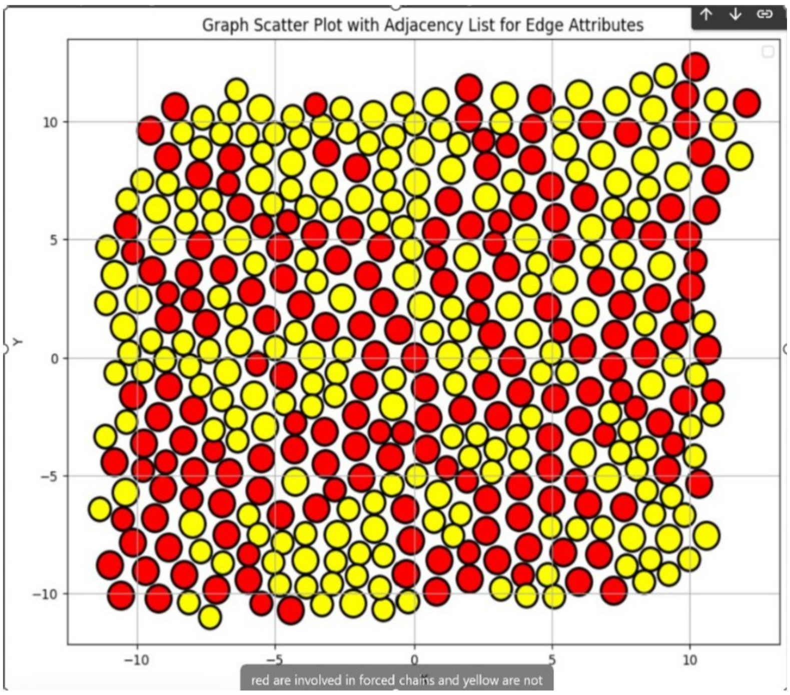


Figure 5: Node positions and labels in the graph

7 Algorithmic Build-Up

- The initial part of the algorithmic build up was to determine a sound algorithm to build the GNN and make data conducive for it.
- The GNN was initially tested with Attention Modules called GAT Conv because:
  a) Dynamic Neighbor Weighting: Transforms node features and computes attention coefficients via a learnable weight vector applied on concatenated node pairs, normalizing scores with soft max to dynamically weigh neighbour contributions.
  b)Multi-Head Aggregation: Utilizes multiple attention heads to capture diverse feature subspaces, aggregating neighbour information with non-linear activations; all parameters are optimized via backpropagation.

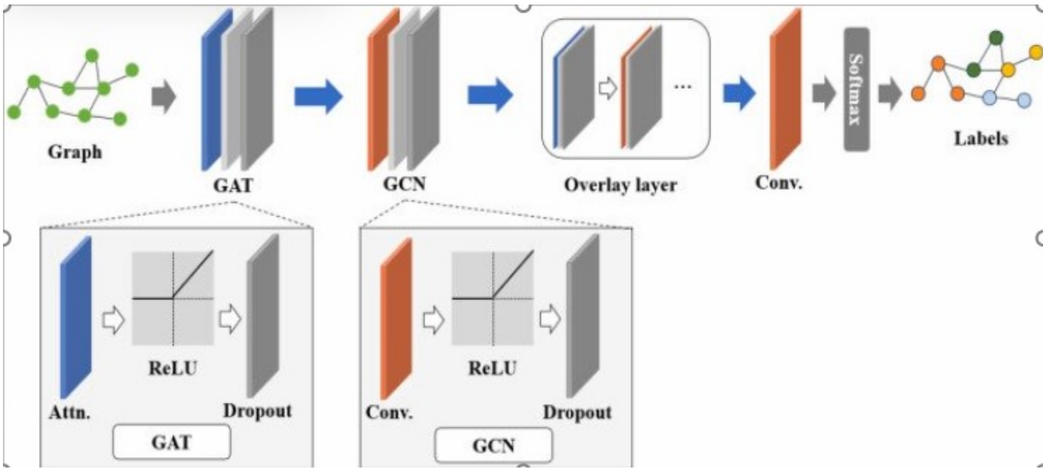


Figure 6: Diagram to explain the GAT architecture

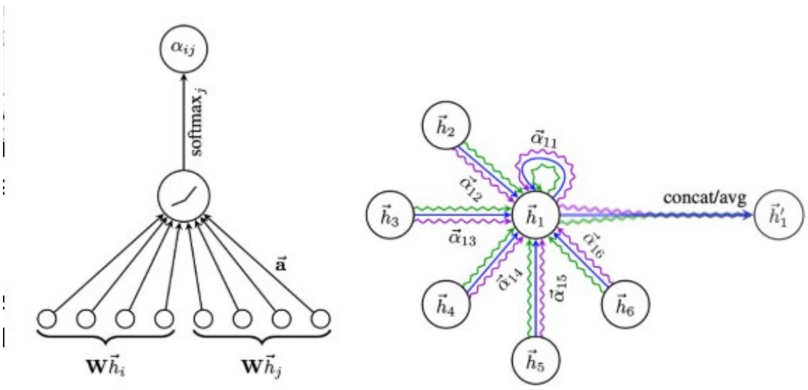


Figure 1: **Left:** The attention mechanism $a(\mathbf{W}\vec{h}_i, \mathbf{W}\vec{h}_j)$ employed by our model, parametrized by a weight vector $\vec{\mathbf{a}} \in \mathbb{R}^{2F'}$, applying a LeakyReLU activation. **Right:** An illustration of multi-head attention (with $K = 3$ heads) by node 1 on its neighborhood. Different arrow styles and colors denote independent attention computations. The aggregated features from each head are concatenated or averaged to obtain $\vec{h}'_1$.

- We produced the intuition that GAT is an overkill when it comes to our problem since we are already working under the assumption of a certain threshold distance and do not want to aggregate information of nodes which are farther from that threshold distance.
- We hence included an edge-encoder layer to ensure that edge features are learnt which brought the training accuracy to 90+ %
- Eventually the problem of Graph Convolution Networks' understanding was complete and the model with correct classification architecture was brought into existence

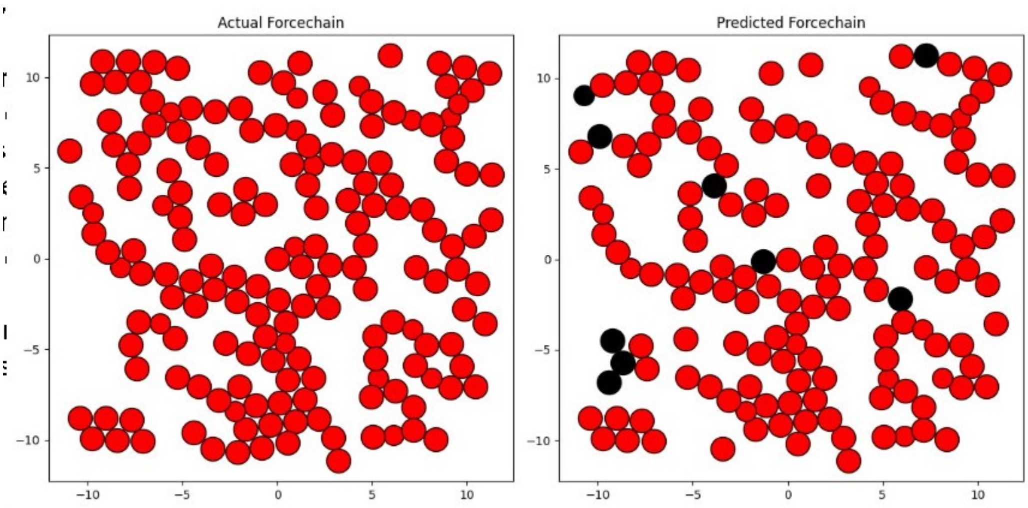


Figure 7: Comparison between actual and predicted force chains

8 Mathematical Background on Voronoi Tessellations

A Voronoi tessellation is a partitioning of space into regions based on the distance to a given set of points. Given a set of seed points $\{p_i\}_{i=1}^N \subset \mathbb{R}^2$, the Voronoi cell V_i corresponding to a seed point p_i is defined as:

$$V_i = \{x \in \mathbb{R}^2 \mid d(x, p_i) \leq d(x, p_j) \quad \forall j \neq i\}, \tag{3}$$

where $d(x, p)$ is the Euclidean distance between points x and p. Each Voronoi cell V_i contains all points that are closer to p_i than to any other seed point.

8.1 Voronoi Tessellations in Our Case

In our problem, the Voronoi tessellations are not general polygons but rather represent **small cylindrical regions** in a packed bed reactor. Each cylinder corresponds to a local fluid domain, where the volume fractions of two immiscible fluids are assigned as node features in our graph representation. These cylinders are packed densely, and the tessellation helps define interaction regions between them.

8.2 Properties of Voronoi Cylinders

- Each Voronoi cell corresponds to a small cylinder, representing a local volume of the multiphase fluid.
- The edges of the Voronoi cells indicate adjacency relationships between cylinders.
- The Delaunay triangulation of the seed points provides a dual representation that can be used for connectivity information.
- The assigned fluid volume fractions, f_1 and f_2, represent the proportions of two immiscible fluids within each cylindrical region.

8.3 Application to Graph Representation

By treating each cylindrical Voronoi region as a graph node and defining edges based on adjacency, we construct a graph neural network (GNN) framework for predicting multiphase flow behavior. Each node has the following attributes:

- Position of the cylinder center (x_i, y_i),

- Edge attributes capturing relative coordinates between cylinders,
- Node features: fluid volume fractions f_1 and f_2.

This graph-based representation allows the GNN to learn spatial and temporal relationships in multiphase flow, facilitating robust predictions of flow evolution.

9 Graph Representation of Multiphase Flow

We represent the multiphase flow domain as a graph $G = (V, E)$, where:

- $V = \{v_i\}$ represents the set of nodes, corresponding to Voronoi tessellations of fluid regions.
- $E = \{e_{ij}\}$ denotes the edges between nodes, representing spatial relationships.
- Each node v_i has features $x_i = (f_i, p_i, v_i)$ where f_i is the fluid fraction, p_i is the position, and v_i is the velocity.

The adjacency matrix A encodes connections between nodes. The edge features e_{ij} may include relative distances and velocity gradients.

The graph dataset is derived from computational fluid dynamics (CFD) simulations. It consists of 100 spatial graphs, each containing 100 timestep graphs. To enhance model generalization, we linearize our GCN and train it to predict the jth graph from any ith graph, ensuring that both graphs belong to the same spatial class.

10 Data Processing

The dataset consists of node coordinates representing the positions of Voronoi tessellation centroids in a cylindrical packed bed. Each edge in the graph is assigned attributes corresponding to the relative edge coordinates, capturing spatial relationships between nodes.

For training, the GNN takes as input a graph representation of the CFD data at a given time step. The output is a vector containing two features:

- f_1: Volume fraction of the first immiscible fluid.
- f_2: Volume fraction of the second immiscible fluid.

These outputs are derived from the Voronoi tessellation of cylinders, ensuring a physically meaningful representation of fluid distribution. The model is trained to predict these values for future time steps, aiding in the accurate simulation of multiphase flow dynamics.

11 Graph Neural Network Model

The Graph Convolutional Network (GCN) update rule is defined as:

$$h_i^{(k+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j^{(k)} + b \right), \tag{4}$$

where:

- $h_i^{(k)}$ is the hidden state of node i at layer k,
- W is a learnable weight matrix,
- α_{ij} represents attention weights if using Graph Attention Networks (GAT),
- b is a bias term,
- σ is a non-linear activation function (e.g., ReLU).

12 Regression Formulation for Multiphase Flow

To predict fluid evolution over time, we treat this as a spatiotemporal regression problem:

$$\hat{y} = f_\theta(G_t), \tag{5}$$

where G_t is the input graph at time t, and $\hat{y}$ is the predicted graph state at $t + \Delta t$.

The loss function used for training is the log-cosh loss, defined as:

$$\mathcal{L} = \sum_i \log\cosh(y_i - \hat{y}_i), \tag{6}$$

which behaves similarly to mean squared error (MSE) but is less sensitive to outliers.

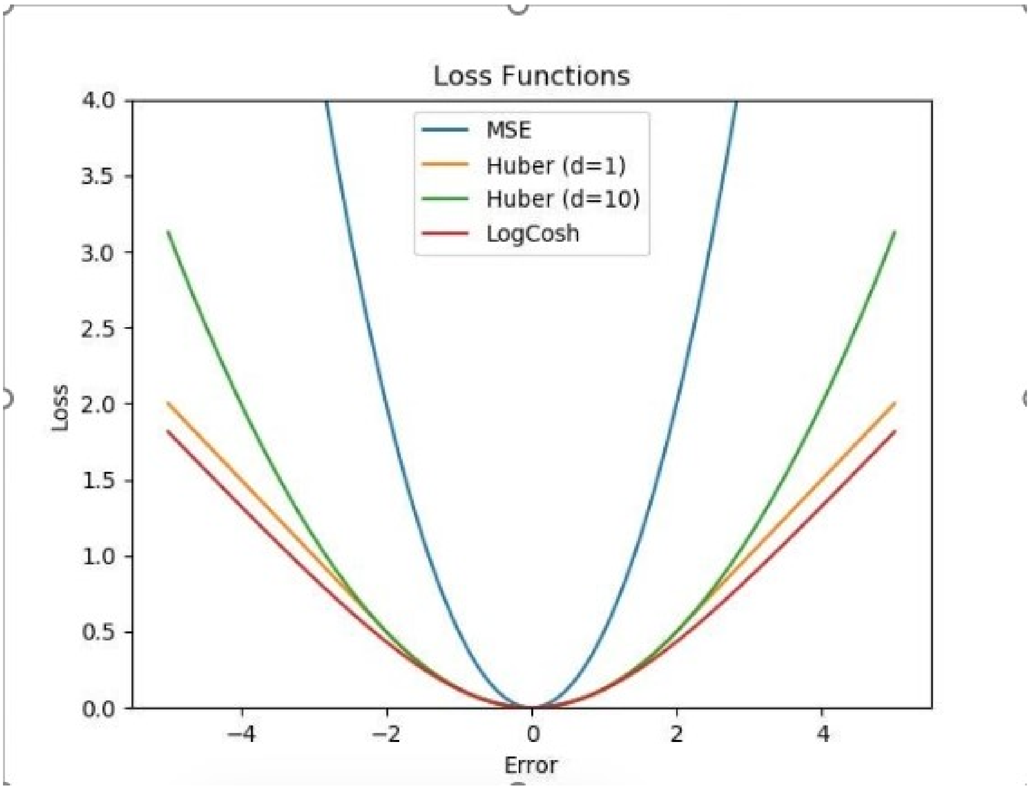


Figure 8: Comparison between different loss functions.

13 Scaling Up to Large Domains

Given a dataset of N graphs, each with T timesteps, the model must generalize over:

- Spatial variations: different geometries of packed beds.
- Temporal evolution: prediction of fluid fraction and velocity over multiple time steps.
- Boundary conditions: periodic constraints applied to maintain physical consistency.

We define the transition function as:

$$G_{t+1} = \mathcal{F}(G_t, \theta), \tag{7}$$

where $\mathcal{F}$ is parameterized by the GNN and θ are the learnable parameters.

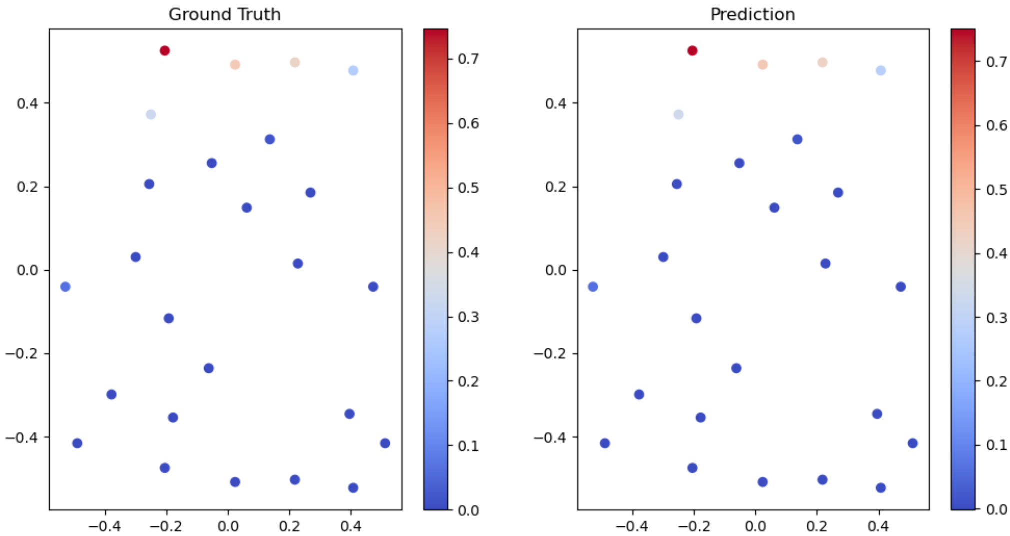


Figure 9: Ground truth vs prediction in the regression problem.

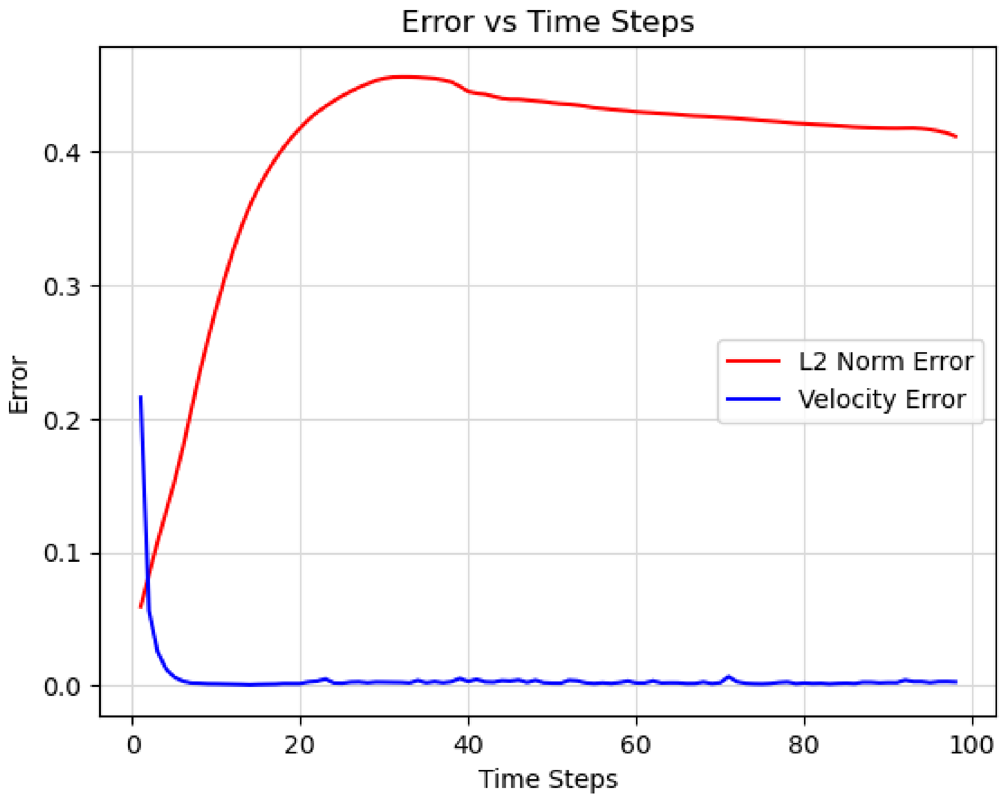


Figure 10: Graph's temporal evolution over time.

14 Conclusion

We developed a graph-based approach to predict multiphase flow evolution using GNNs. By leveraging Voronoi tessellations, attention mechanisms, and advanced loss functions, we improve the robustness and generalizability of our model for large-scale fluid simulations.

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